Random Variables, Expectation and Joint Distributions

Part I: Random Vectors and Joint Distributions

Problem 1

Let (X, Y) be a discrete random vector with joint PMF

$$\mathrm{P}_{XY}(x, y) = \begin{cases} \dfrac{x + 2y}{18}, & x, y \in \{1, 2\}, \\ 0, & \text{otherwise.} \end{cases}$$

(i) Verify that P_{XY} is a valid joint PMF.

(ii) Find the marginal PMFs P_X and P_Y.

(iii) Find $\mathrm{P}_{Y|X}(y \mid 1)$ and $\mathrm{P}_{Y|X}(y \mid 2)$.

(iv) Determine whether X and Y are independent.

Solution.

The four possible values of the joint PMF are

$Y \backslash X$	1	2
1	$\frac{3}{18}$	$\frac{4}{18}$
2	$\frac{5}{18}$	$\frac{6}{18}$

(i) Verification.

All four probabilities are nonnegative, and

$$\frac{3 + 4 + 5 + 6}{18} = \frac{18}{18} = 1.$$

Hence P_{XY} is a valid joint PMF.

(ii) Marginal PMFs.

For X,

$$\mathrm{P}_X(1) = \frac{3}{18} + \frac{5}{18} = \frac{4}{9},$$
$$\mathrm{P}_X(2) = \frac{4}{18} + \frac{6}{18} = \frac{5}{9}.$$

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Thus,

$$\mathrm{P}_X(x) = \begin{cases} \frac{4}{9}, & x = 1, \\ \frac{5}{9}, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

For Y,

$$\mathrm{P}_Y(1) = \frac{3}{18} + \frac{4}{18} = \frac{7}{18},$$
$$\mathrm{P}_Y(2) = \frac{5}{18} + \frac{6}{18} = \frac{11}{18}.$$

Hence,

$$\mathrm{P}_Y(y) = \begin{cases} \frac{7}{18}, & y = 1, \\ \frac{11}{18}, & y = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(iii) Conditional PMFs.

For $X = 1$,

$$\mathrm{P}_{Y|X}(y \mid 1) = \frac{\mathrm{P}_{XY}(1, y)}{\mathrm{P}_X(1)}.$$

Therefore,

$$\mathrm{P}_{Y|X}(1 \mid 1) = \frac{3/18}{4/9} = \frac{3}{8},$$

$$\mathrm{P}_{Y|X}(2 \mid 1) = \frac{5/18}{4/9} = \frac{5}{8}.$$

Thus,

$$\mathrm{P}_{Y|X}(y \mid 1) = \begin{cases} \frac{3}{8}, & y = 1, \\ \frac{5}{8}, & y = 2, \\ 0, & \text{otherwise.} \end{cases}$$

For $X = 2$,

$$\mathrm{P}_{Y|X}(1 \mid 2) = \frac{4/18}{5/9} = \frac{2}{5},$$

$$\mathrm{P}_{Y|X}(2 \mid 2) = \frac{6/18}{5/9} = \frac{3}{5}.$$

Hence,

$$\mathrm{P}_{Y|X}(y \mid 2) = \begin{cases} \frac{2}{5}, & y = 1, \\ \frac{3}{5}, & y = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(iv) Independence.

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If X and Y were independent, we would have

$$\mathrm{P}_{XY}(1,1) = \mathrm{P}_X(1)\mathrm{P}_Y(1).$$

But

$$\mathrm{P}_{XY}(1,1) = \frac{1}{6},$$

whereas

$$\mathrm{P}_X(1)\mathrm{P}_Y(1) = \frac{4}{9}\frac{7}{18} = \frac{14}{81}.$$

Since

$$\frac{1}{6} \neq \frac{14}{81},$$

X and Y are not independent.

$$\boxed{X \text{ and } Y \text{ are not independent.}}$$

Problem 2

Let X and Y be random variables satisfying

$$\mathrm{P}(X \in \{0,1\}) = \mathrm{P}(Y \in \{0,1\}) = 1.$$

Suppose that

$$\mathrm{P}(X = 1, Y = 1) = \mathrm{P}(X = 1)\mathrm{P}(Y = 1).$$

Show that

$$\mathrm{P}_{XY}(x,y) = \mathrm{P}_X(x)\mathrm{P}_Y(y)$$

for every $x, y \in \{0,1\}$. Hence prove that X and Y are independent.

Solution.

Let

$$p = \mathrm{P}(X = 1), \qquad q = \mathrm{P}(Y = 1).$$

The given condition is

$$\mathrm{P}(X = 1, Y = 1) = pq.$$

Thus,

$$\mathrm{P}_{XY}(1,1) = pq = \mathrm{P}_X(1)\mathrm{P}_Y(1).$$

Next,

$$\begin{aligned}\mathrm{P}_{XY}(1,0) &= \mathrm{P}(X = 1) - \mathrm{P}(X = 1, Y = 1)\\ &= p - pq\\ &= p(1-q)\\ &= \mathrm{P}_X(1)\mathrm{P}_Y(0).\end{aligned}$$

Similarly,

$$\begin{aligned}\mathrm{P}_{XY}(0,1) &= \mathrm{P}(Y = 1) - \mathrm{P}(X = 1, Y = 1)\\ &= q - pq\\ &= (1-p)q\\ &= \mathrm{P}_X(0)\mathrm{P}_Y(1).\end{aligned}$$

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Finally,

$$\begin{aligned} \mathrm{P}_{XY}(0,0) &= 1 - \mathrm{P}_{XY}(1,1) - \mathrm{P}_{XY}(1,0) - \mathrm{P}_{XY}(0,1) \\ &= 1 - pq - p(1-q) - (1-p)q \\ &= 1 - p - q + pq \\ &= (1-p)(1-q) \\ &= \mathrm{P}_X(0)\mathrm{P}_Y(0). \end{aligned}$$

Therefore,

$$\mathrm{P}_{XY}(x,y) = \mathrm{P}_X(x)\mathrm{P}_Y(y)$$

for every $x, y \in \{0,1\}$. Hence,

$$\boxed{X \text{ and } Y \text{ are independent.}}$$

Problem 3

A sample of size 3 is drawn with replacement from an urn containing 3 white balls, 2 black balls, and 3 red balls.

Let

$$X_1 = \text{number of white balls in the sample,}$$

and

$$X_2 = \text{number of black balls in the sample.}$$

Find the joint PMF $\mathrm{P}_{X_1,X_2}(x_1,x_2)$ and the marginal PMFs P_{X_1} and P_{X_2}. Determine whether X_1 and X_2 are independent.

Solution.

On each draw,

$$\mathrm{P}(W) = \frac{3}{8}, \qquad \mathrm{P}(B) = \frac{2}{8} = \frac{1}{4}, \qquad \mathrm{P}(R) = \frac{3}{8}.$$

If $X_1 = x_1$ and $X_2 = x_2$, then the number of red balls is

$$3 - x_1 - x_2.$$

Therefore, by the multinomial formula,

$$\mathrm{P}_{X_1,X_2}(x_1,x_2) = \frac{3!}{x_1!x_2!(3-x_1-x_2)!}\left(\frac{3}{8}\right)^{x_1}\left(\frac{1}{4}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2},$$

for

$$x_1 = 0,1,2,3, \qquad x_2 = 0,1,2,3, \qquad x_1 + x_2 \le 3,$$

and it is zero otherwise.

For the marginal distribution of X_1, each draw is either white or not white. Hence

$$X_1 \sim \mathrm{Bin}\left(3, \frac{3}{8}\right),$$

so

$$\mathrm{P}_{X_1}(x_1) = \binom{3}{x_1}\left(\frac{3}{8}\right)^{x_1}\left(\frac{5}{8}\right)^{3-x_1}, \qquad x_1 = 0,1,2,3.$$

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Similarly,

$$X_2 \sim \text{Bin}\left(3, \frac{1}{4}\right),$$

so

$$\text{P}_{X_2}(x_2) = \binom{3}{x_2}\left(\frac{1}{4}\right)^{x_2}\left(\frac{3}{4}\right)^{3-x_2}, \qquad x_2 = 0, 1, 2, 3.$$

To test independence, consider $X_1 = 3$ and $X_2 = 1$. Since only three balls are drawn,

$$\text{P}_{X_1,X_2}(3, 1) = 0.$$

However,

$$\text{P}_{X_1}(3) = \left(\frac{3}{8}\right)^3 > 0$$

and

$$\text{P}_{X_2}(1) = \binom{3}{1}\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^2 > 0.$$

Thus,

$$\text{P}_{X_1,X_2}(3, 1) \neq \text{P}_{X_1}(3)\text{P}_{X_2}(1).$$

Therefore,

$$\boxed{X_1 \text{ and } X_2 \text{ are not independent.}}$$

Derivation:

The marginal PMF of X_1 is obtained by summing the joint PMF over all possible values of x_2:

$$P_{X_1}(x_1) = \sum_{x_2=0}^{3-x_1} P_{X_1,X_2}(x_1, x_2).$$

Using the joint PMF,

$$P_{X_1,X_2}(x_1, x_2) = \frac{3!}{x_1!x_2!(3-x_1-x_2)!}\left(\frac{3}{8}\right)^{x_1}\left(\frac{2}{8}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2},$$

we obtain

$$P_{X_1}(x_1) = \sum_{x_2=0}^{3-x_1} \frac{3!}{x_1!x_2!(3-x_1-x_2)!}\left(\frac{3}{8}\right)^{x_1}\left(\frac{2}{8}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2}.$$

Taking the terms independent of x_2 outside the summation,

$$P_{X_1}(x_1) = \frac{3!}{x_1!}\left(\frac{3}{8}\right)^{x_1}\sum_{x_2=0}^{3-x_1} \frac{1}{x_2!(3-x_1-x_2)!}\left(\frac{2}{8}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2}.$$

Multiplying and dividing by $(3-x_1)!$,

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$$P_{X_1}(x_1) = \frac{3!}{x_1!(3-x_1)!}\left(\frac{3}{8}\right)^{x_1}\sum_{x_2=0}^{3-x_1}\frac{(3-x_1)!}{x_2!(3-x_1-x_2)!}\left(\frac{2}{8}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2}.$$

Therefore,

$$P_{X_1}(x_1) = \binom{3}{x_1}\left(\frac{3}{8}\right)^{x_1}\sum_{x_2=0}^{3-x_1}\binom{3-x_1}{x_2}\left(\frac{2}{8}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2}.$$

Using the binomial theorem,

$$\sum_{k=0}^{n}\binom{n}{k}a^k b^{n-k} = (a+b)^n,$$

we get

$$\sum_{x_2=0}^{3-x_1}\binom{3-x_1}{x_2}\left(\frac{2}{8}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2} = \left(\frac{2}{8}+\frac{3}{8}\right)^{3-x_1}.$$

Hence,

$$\boxed{P_{X_1}(x_1) = \binom{3}{x_1}\left(\frac{3}{8}\right)^{x_1}\left(\frac{5}{8}\right)^{3-x_1}}, \qquad x_1 = 0,1,2,3.$$

Thus,

$$\boxed{X_1 \sim \text{Bin}\left(3, \frac{3}{8}\right).}$$

Problem 4

The joint PDF of (X, Y) is

$$f_{XY}(x,y) = \begin{cases} 4xy, & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(i) Find the marginal PDFs f_X and f_Y.

(ii) Determine whether X and Y are independent.

(iii) Calculate

$$\text{P}\left(0 < X < \frac{1}{2},\ \frac{1}{4} < Y < 1\right).$$

(iv) Calculate $\text{P}(X + Y < 1)$.

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Solution.

(i) Marginal PDFs.

For $0 < x < 1$,

$$\begin{aligned} f_X(x) &= \int_0^1 4xy\,dy \\ &= 4x\left[\frac{y^2}{2}\right]_0^1 \\ &= 2x. \end{aligned}$$

Thus,

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Similarly, for $0 < y < 1$,

$$\begin{aligned} f_Y(y) &= \int_0^1 4xy\,dx \\ &= 4y\left[\frac{x^2}{2}\right]_0^1 \\ &= 2y. \end{aligned}$$

Hence,

$$f_Y(y) = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(ii) Independence.

For $0 < x < 1$ and $0 < y < 1$,

$$f_X(x)f_Y(y) = (2x)(2y) = 4xy = f_{XY}(x, y).$$

Hence,

$$\boxed{X \text{ and } Y \text{ are independent.}}$$

(iii) Since the region is a rectangle,

$$\mathrm{P}\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < 1\right) = \int_0^{1/2}\int_{1/4}^1 4xy\,dy\,dx.$$

First integrate with respect to y:

$$\int_{1/4}^1 4xy\,dy = 2x\left(1 - \frac{1}{16}\right) = \frac{15x}{8}.$$

Therefore,

$$\begin{aligned} \mathrm{P}\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < 1\right) &= \int_0^{1/2} \frac{15x}{8}\,dx \\ &= \frac{15}{8}\left[\frac{x^2}{2}\right]_0^{1/2} \\ &= \frac{15}{64}. \end{aligned}$$

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Thus,

$$\boxed{\mathrm{P}\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < 1\right) = \frac{15}{64}.}$$

(iv) The event $X + Y < 1$ corresponds to

$$0 < x < 1, \qquad 0 < y < 1 - x.$$

Hence,

$$\begin{aligned}
\mathrm{P}(X + Y < 1) &= \int_0^1 \int_0^{1-x} 4xy \, dy \, dx \\
&= \int_0^1 2x(1-x)^2 \, dx \\
&= 2\int_0^1 (x - 2x^2 + x^3) \, dx \\
&= 2\left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right]_0^1 \\
&= 2\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) \\
&= \frac{1}{6}.
\end{aligned}$$

Therefore,

$$\boxed{\mathrm{P}(X + Y < 1) = \frac{1}{6}.}$$

Problem 5

The joint PDF of (X, Y) is

$$f_{XY}(x, y) = \begin{cases} ce^{-(2x+3y)}, & 0 < x < y < \infty, \\ 0, & \text{otherwise}, \end{cases}$$

where c is a constant.

(i) Find c.

(ii) Find the marginal PDFs f_X and f_Y.

(iii) Determine whether X and Y are independent.

(iv) Calculate $\mathrm{P}(X < Y/2)$.

Solution.

(i) Finding c.

Since the total probability is 1,

$$\int_0^\infty \int_x^\infty ce^{-(2x+3y)} \, dy \, dx = 1.$$

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Integrating first with respect to y,

$$\int_x^\infty e^{-3y}\,dy = \frac{1}{3}e^{-3x}.$$

Therefore,

$$\begin{aligned} 1 &= \frac{c}{3}\int_0^\infty e^{-5x}\,dx \\ &= \frac{c}{3}\left(\frac{1}{5}\right) \\ &= \frac{c}{15}. \end{aligned}$$

Hence,

$$\boxed{c = 15.}$$

(ii) Marginal PDFs.

For $x > 0$, the support requires $y > x$. Therefore,

$$\begin{aligned} f_X(x) &= \int_x^\infty 15e^{-(2x+3y)}\,dy \\ &= 15e^{-2x}\int_x^\infty e^{-3y}\,dy \\ &= 15e^{-2x}\left(\frac{1}{3}e^{-3x}\right) \\ &= 5e^{-5x}. \end{aligned}$$

Thus,

$$f_X(x) = \begin{cases} 5e^{-5x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

For $y > 0$, the support requires $0 < x < y$. Thus,

$$\begin{aligned} f_Y(y) &= \int_0^y 15e^{-(2x+3y)}\,dx \\ &= 15e^{-3y}\int_0^y e^{-2x}\,dx \\ &= 15e^{-3y}\left[\frac{1-e^{-2y}}{2}\right] \\ &= \frac{15}{2}\left(e^{-3y} - e^{-5y}\right). \end{aligned}$$

Hence,

$$f_Y(y) = \begin{cases} \dfrac{15}{2}\left(e^{-3y} - e^{-5y}\right), & y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

(iii) Independence.

The joint density is positive only on the triangular region

$$0 < x < y.$$

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The marginal densities are positive for every $x > 0$ and every $y > 0$. Thus there are points with $x \geq y > 0$ for which

$$f_X(x) f_Y(y) > 0$$

but

$$f_{XY}(x, y) = 0.$$

Therefore,

$$\boxed{X \text{ and } Y \text{ are not independent.}}$$

(iv) The event

$$X < \frac{Y}{2}$$

is equivalent to

$$Y > 2X.$$

Since $Y > 2X$ already implies $Y > X$, the required probability is

$$\mathrm{P}\left(X < \frac{Y}{2}\right) = \int_0^\infty \int_{2x}^\infty 15e^{-(2x+3y)}\, dy\, dx.$$

Integrating with respect to y,

$$\begin{aligned}
\mathrm{P}\left(X < \frac{Y}{2}\right) &= \int_0^\infty 15e^{-2x} \left(\frac{1}{3}e^{-6x}\right) dx \\
&= 5\int_0^\infty e^{-8x}\, dx \\
&= 5\left(\frac{1}{8}\right) \\
&= \frac{5}{8}.
\end{aligned}$$

Hence,

$$\boxed{\mathrm{P}\left(X < \frac{Y}{2}\right) = \frac{5}{8}.}$$

Problem 6

The joint PMF of (X, Y) is given by

$Y \backslash X$	1	2	3	4
4	0.08	0.11	0.09	0.03
5	0.04	0.12	0.21	0.05
6	0.09	0.06	0.08	0.04

(i) Find the marginal PMFs P_X and P_Y.

(ii) Find the conditional PMF $\mathrm{P}_{X|Y}(x \mid 5)$.

(iii) Calculate $\mathrm{P}(X + Y \leq 8)$ and $\mathrm{P}(X + Y > 7)$.

(iv) Calculate $\mathrm{P}(XY \leq 14)$ and $\mathrm{P}(XY > 18)$.

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(v) Find $\mathrm{P}(X = 3 \mid Y = 5)$ and $\mathrm{P}(Y = 5 \mid X = 3)$.

Solution.

(i) Marginal PMFs.

Sum each column to obtain P_X:

$$\begin{aligned}
\mathrm{P}_X(1) &= 0.08 + 0.04 + 0.09 = 0.21,\\
\mathrm{P}_X(2) &= 0.11 + 0.12 + 0.06 = 0.29,\\
\mathrm{P}_X(3) &= 0.09 + 0.21 + 0.08 = 0.38,\\
\mathrm{P}_X(4) &= 0.03 + 0.05 + 0.04 = 0.12.
\end{aligned}$$

Thus,

x	1	2	3	4
$\mathrm{P}_X(x)$	0.21	0.29	0.38	0.12

Sum each row to obtain P_Y:

$$\begin{aligned}
\mathrm{P}_Y(4) &= 0.08 + 0.11 + 0.09 + 0.03 = 0.31,\\
\mathrm{P}_Y(5) &= 0.04 + 0.12 + 0.21 + 0.05 = 0.42,\\
\mathrm{P}_Y(6) &= 0.09 + 0.06 + 0.08 + 0.04 = 0.27.
\end{aligned}$$

Hence,

y	4	5	6
$\mathrm{P}_Y(y)$	0.31	0.42	0.27

(ii) Conditional PMF $\mathrm{P}_{X|Y}(x \mid 5)$.

$$\mathrm{P}_{X|Y}(x \mid 5) = \frac{\mathrm{P}_{XY}(x, 5)}{\mathrm{P}_Y(5)}.$$

Since $\mathrm{P}_Y(5) = 0.42$,

$$\begin{aligned}
\mathrm{P}_{X|Y}(1 \mid 5) &= \frac{0.04}{0.42} = \frac{2}{21},\\
\mathrm{P}_{X|Y}(2 \mid 5) &= \frac{0.12}{0.42} = \frac{2}{7},\\
\mathrm{P}_{X|Y}(3 \mid 5) &= \frac{0.21}{0.42} = \frac{1}{2},\\
\mathrm{P}_{X|Y}(4 \mid 5) &= \frac{0.05}{0.42} = \frac{5}{42}.
\end{aligned}$$

Thus,

$$\mathrm{P}_{X|Y}(x \mid 5) = \begin{cases} \frac{2}{21}, & x = 1,\\ \frac{2}{7}, & x = 2,\\ \frac{1}{2}, & x = 3,\\ \frac{5}{42}, & x = 4,\\ 0, & \text{otherwise.} \end{cases}$$

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(iii) Probabilities involving $X+Y$.

For $X+Y \leq 8$:

- If $Y=4$, then $X=1,2,3,4$ are allowed.
- If $Y=5$, then $X=1,2,3$ are allowed.
- If $Y=6$, then $X=1,2$ are allowed.

Therefore,

$$\begin{aligned}\mathrm{P}(X+Y\leq 8) &= (0.08+0.11+0.09+0.03)\\ &\quad +(0.04+0.12+0.21)\\ &\quad +(0.09+0.06)\\ &= 0.31+0.37+0.15\\ &= 0.83.\end{aligned}$$

For $X+Y>7$:

- If $Y=4$, only $X=4$ is allowed.
- If $Y=5$, $X=3,4$ are allowed.
- If $Y=6$, $X=2,3,4$ are allowed.

Thus,

$$\begin{aligned}\mathrm{P}(X+Y>7) &= 0.03+(0.21+0.05)+(0.06+0.08+0.04)\\ &= 0.03+0.26+0.18\\ &= 0.47.\end{aligned}$$

Hence,

$$\boxed{\mathrm{P}(X+Y\leq 8)=0.83, \qquad \mathrm{P}(X+Y>7)=0.47.}$$

(iv) Probabilities involving XY.

For $XY \leq 14$:

- If $Y=4$, $X=1,2,3$ are allowed.
- If $Y=5$, $X=1,2$ are allowed.
- If $Y=6$, $X=1,2$ are allowed.

Therefore,

$$\begin{aligned}\mathrm{P}(XY\leq 14) &= (0.08+0.11+0.09)+(0.04+0.12)+(0.09+0.06)\\ &= 0.28+0.16+0.15\\ &= 0.59.\end{aligned}$$

For $XY>18$:

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- For $Y = 4$, no value of X gives $XY > 18$.
- For $Y = 5$, only $X = 4$ gives $XY > 18$.
- For $Y = 6$, only $X = 4$ gives $XY > 18$; $X = 3$ gives exactly 18.

Hence,

$$\mathrm{P}(XY > 18) = 0.05 + 0.04 = 0.09.$$

Therefore,

$$\boxed{\mathrm{P}(XY \leq 14) = 0.59, \qquad \mathrm{P}(XY > 18) = 0.09.}$$

(v) Conditional probabilities.

First,

$$\begin{aligned}\mathrm{P}(X = 3 \mid Y = 5) &= \frac{\mathrm{P}(X = 3, Y = 5)}{\mathrm{P}(Y = 5)} \\ &= \frac{0.21}{0.42} \\ &= \frac{1}{2}.\end{aligned}$$

Next,

$$\begin{aligned}\mathrm{P}(Y = 5 \mid X = 3) &= \frac{\mathrm{P}(X = 3, Y = 5)}{\mathrm{P}(X = 3)} \\ &= \frac{0.21}{0.38} \\ &= \frac{21}{38}.\end{aligned}$$

Thus,

$$\boxed{\mathrm{P}(X = 3 \mid Y = 5) = \frac{1}{2}, \qquad \mathrm{P}(Y = 5 \mid X = 3) = \frac{21}{38}.}$$

Part II: Expectation and Variance of Random Variables

Problem 7

Let X be a discrete random variable with PMF

$$\mathrm{P}_X(x) = \begin{cases} \dfrac{1}{n}, & x \in \{1, 2, \ldots, n\}, \\ 0, & \text{otherwise}, \end{cases}$$

where $n \geq 2$ is an integer. Find $\mathrm{E}[X]$ and $\mathrm{Var}(X)$.

Solution.

$$\mathrm{E}[X] = \sum_{x=1}^{n} x \, \mathrm{P}_X(x).$$

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Since $\mathrm{P}_X(x) = 1/n$,

$$\mathrm{E}[X] = \frac{1}{n}\sum_{x=1}^{n} x.$$

Using

$$\sum_{x=1}^{n} x = \frac{n(n+1)}{2},$$

we obtain

$$\mathrm{E}[X] = \frac{1}{n}\frac{n(n+1)}{2} = \frac{n+1}{2}.$$

Next,

$$\mathrm{E}[X^2] = \sum_{x=1}^{n} x^2 \mathrm{P}_X(x) = \frac{1}{n}\sum_{x=1}^{n} x^2.$$

Using

$$\sum_{x=1}^{n} x^2 = \frac{n(n+1)(2n+1)}{6},$$

we have

$$\mathrm{E}[X^2] = \frac{(n+1)(2n+1)}{6}.$$

Therefore,

$$\begin{aligned}\mathrm{Var}(X) &= \mathrm{E}[X^2] - \Big(\mathrm{E}[X]\Big)^2 \\ &= \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 \\ &= \frac{n^2-1}{12}.\end{aligned}$$

Hence,

$$\boxed{\mathrm{E}[X] = \frac{n+1}{2}, \qquad \mathrm{Var}(X) = \frac{n^2-1}{12}.}$$

Problem 8

A fair coin is tossed independently three times. Let X denote the number of tails obtained, and define

$$Y = X^2, \qquad Z = 2X^2 + 1.$$

Find $\mathrm{E}[Y]$, $\mathrm{Var}(Y)$, $\mathrm{E}[Z]$, and $\mathrm{Var}(Z)$.

Solution.

Since X is the number of tails in three independent fair tosses,

$$\mathrm{P}(X = x) = \binom{3}{x}\left(\frac{1}{2}\right)^3, \qquad x = 0, 1, 2, 3.$$

Thus,

x	0	1	2	3
$\mathrm{P}_X(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

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Since $Y = X^2$,

$$\mathrm{E}[Y] = \mathrm{E}[X^2] = \sum_{x=0}^{3} x^2 \mathrm{P}_X(x).$$

Hence,

$$\begin{aligned}\mathrm{E}[Y] &= 0^2\frac{1}{8} + 1^2\frac{3}{8} + 2^2\frac{3}{8} + 3^2\frac{1}{8} \\ &= \frac{3+12+9}{8} = 3.\end{aligned}$$

To find $\mathrm{Var}(Y)$, first compute $\mathrm{E}[Y^2]$:

$$\mathrm{E}[Y^2] = \mathrm{E}[X^4] = \sum_{x=0}^{3} x^4 \mathrm{P}_X(x).$$

Therefore,

$$\begin{aligned}\mathrm{E}[Y^2] &= 0^4\frac{1}{8} + 1^4\frac{3}{8} + 2^4\frac{3}{8} + 3^4\frac{1}{8} \\ &= \frac{3+48+81}{8} = \frac{33}{2}.\end{aligned}$$

Thus,

$$\begin{aligned}\mathrm{Var}(Y) &= \mathrm{E}[Y^2] - \Big(\mathrm{E}[Y]\Big)^2 \\ &= \frac{33}{2} - 9 = \frac{15}{2}.\end{aligned}$$

Since $Z = 2Y + 1$, linearity of expectation gives

$$\mathrm{E}[Z] = 2\mathrm{E}[Y] + 1 = 2(3) + 1 = 7.$$

Also,

$$\mathrm{Var}(Z) = \mathrm{Var}(2Y+1) = 4\,\mathrm{Var}(Y) = 4\left(\frac{15}{2}\right) = 30.$$

Hence,

$$\boxed{\mathrm{E}[Y] = 3, \quad \mathrm{Var}(Y) = \frac{15}{2}, \quad \mathrm{E}[Z] = 7, \quad \mathrm{Var}(Z) = 30.}$$

Problem 9

A fair die is thrown repeatedly until a 6 is obtained. Let X denote the number of throws required to obtain the first 6.

Find the PMF P_X and hence calculate $\mathrm{E}[X]$.

Solution.

For $X = x$, the first $x - 1$ throws must be different from 6 and the xth throw must be 6. Therefore,

$$\mathrm{P}(X = x) = \left(\frac{5}{6}\right)^{x-1}\left(\frac{1}{6}\right), \qquad x = 1, 2, \ldots$$

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Thus,

$$\mathrm{P}_X(x) = \begin{cases} \dfrac{1}{6}\left(\dfrac{5}{6}\right)^{x-1}, & x = 1, 2, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

Now,

$$\mathrm{E}[X] = \sum_{x=1}^{\infty} x\,\mathrm{P}_X(x) = \frac{1}{6}\sum_{x=1}^{\infty} x\left(\frac{5}{6}\right)^{x-1}.$$

Using

$$\sum_{x=1}^{\infty} x r^{x-1} = \frac{1}{(1-r)^2}, \qquad |r| < 1,$$

with $r = 5/6$,

$$\begin{aligned} \mathrm{E}[X] &= \frac{1}{6}\frac{1}{(1-5/6)^2} \\ &= \frac{1}{6}\frac{1}{(1/6)^2} \\ &= 6. \end{aligned}$$

Hence,

$$\boxed{\mathrm{P}_X(x) = \frac{1}{6}\left(\frac{5}{6}\right)^{x-1},\ x = 1, 2, \ldots, \qquad \mathrm{E}[X] = 6.}$$

Problem 10

Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} \dfrac{x}{2}, & 0 < x \le 1, \\ \dfrac{1}{2}, & 1 < x \le 2, \\ \dfrac{3-x}{2}, & 2 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Define

$$Y = X^2 - 5X + 3.$$

Without finding the PDF of Y, calculate $\mathrm{E}[Y]$.

Solution.

By linearity of expectation,

$$\mathrm{E}[Y] = \mathrm{E}[X^2] - 5\mathrm{E}[X] + 3.$$

We first calculate $\mathrm{E}[X]$:

$$\begin{aligned} \mathrm{E}[X] &= \int_{-\infty}^{\infty} x f_X(x)\,dx \\ &= \int_0^1 x\left(\frac{x}{2}\right)dx + \int_1^2 x\left(\frac{1}{2}\right)dx + \int_2^3 x\left(\frac{3-x}{2}\right)dx. \end{aligned}$$

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The three terms are

$$\int_0^1 \frac{x^2}{2}\,dx = \frac{1}{6},$$
$$\int_1^2 \frac{x}{2}\,dx = \frac{3}{4},$$

and

$$\int_2^3 \frac{x(3-x)}{2}\,dx = \frac{7}{12}.$$

Therefore,

$$\mathrm{E}[X] = \frac{1}{6} + \frac{3}{4} + \frac{7}{12} = \frac{3}{2}.$$

Next,

$$\begin{aligned}\mathrm{E}[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x)\,dx \\ &= \int_0^1 x^2\left(\frac{x}{2}\right)dx + \int_1^2 x^2\left(\frac{1}{2}\right)dx + \int_2^3 x^2\left(\frac{3-x}{2}\right)dx.\end{aligned}$$

The three terms are

$$\int_0^1 \frac{x^3}{2}\,dx = \frac{1}{8},$$
$$\int_1^2 \frac{x^2}{2}\,dx = \frac{7}{6},$$

and

$$\int_2^3 \frac{x^2(3-x)}{2}\,dx = \frac{11}{8}.$$

Thus,

$$\mathrm{E}[X^2] = \frac{1}{8} + \frac{7}{6} + \frac{11}{8} = \frac{8}{3}.$$

Hence,

$$\begin{aligned}\mathrm{E}[Y] &= \frac{8}{3} - 5\left(\frac{3}{2}\right) + 3 \\ &= \frac{8}{3} - \frac{15}{2} + 3 \\ &= -\frac{11}{6}.\end{aligned}$$

Therefore,

$$\boxed{\mathrm{E}[Y] = -\frac{11}{6}.}$$

Problem 11

A target consists of three concentric circles with radii

$$\frac{1}{\sqrt{3}}, \qquad 1, \qquad \sqrt{3}.$$

A shot earns 4 points if it falls inside the innermost circle, 3 points if it falls in the next ring, 2 points if it falls in the outermost ring, and 0 points if it falls outside the target.

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Let X denote the distance of the hit from the centre. Suppose that

$$f_X(x) = \begin{cases} \dfrac{2}{\pi(1+x^2)}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

If S denotes the score obtained from one shot, find $\mathrm{E}[S]$.

Solution.

The score S is determined by the value of X:

$$S = \begin{cases} 4, & 0 < X < \dfrac{1}{\sqrt{3}}, \\ 3, & \dfrac{1}{\sqrt{3}} \le X < 1, \\ 2, & 1 \le X < \sqrt{3}, \\ 0, & X \ge \sqrt{3}. \end{cases}$$

For $a > 0$,

$$\mathrm{P}(X < a) = \int_0^a \frac{2}{\pi(1+x^2)}\,dx = \frac{2}{\pi}\tan^{-1}(a).$$

Thus,

$$\mathrm{P}\left(X < \frac{1}{\sqrt{3}}\right) = \frac{2}{\pi}\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{2}{\pi}\frac{\pi}{6} = \frac{1}{3}.$$

Also,

$$\mathrm{P}(X < 1) = \frac{2}{\pi}\tan^{-1}(1) = \frac{1}{2},$$

and

$$\mathrm{P}(X < \sqrt{3}) = \frac{2}{\pi}\tan^{-1}(\sqrt{3}) = \frac{2}{3}.$$

Therefore,

$$\mathrm{P}(S = 4) = \frac{1}{3},$$
$$\mathrm{P}(S = 3) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6},$$
$$\mathrm{P}(S = 2) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6},$$

and

$$\mathrm{P}(S = 0) = 1 - \frac{2}{3} = \frac{1}{3}.$$

Hence,

$$\begin{aligned} \mathrm{E}[S] &= 4\mathrm{P}(S=4) + 3\mathrm{P}(S=3) + 2\mathrm{P}(S=2) + 0\mathrm{P}(S=0) \\ &= 4\left(\frac{1}{3}\right) + 3\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) \\ &= \frac{13}{6}. \end{aligned}$$

Therefore,

$$\boxed{\mathrm{E}[S] = \frac{13}{6}.}$$

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Problem 12

Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 6x(1-x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(i) Verify that f_X is a valid PDF.

(ii) Calculate $\mathrm{E}[X]$.

(iii) Calculate $\mathrm{E}[X^2]$.

(iv) Hence find $\mathrm{Var}(X)$ and the standard deviation of X.

Solution.

(i) Verification of the PDF.

For $0 < x < 1$,

$$6x(1-x) \geq 0.$$

Also,

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x)\,dx &= \int_0^1 6x(1-x)\,dx \\ &= 6\int_0^1 (x - x^2)\,dx \\ &= 6\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 \\ &= 6\left(\frac{1}{2} - \frac{1}{3}\right) = 1. \end{aligned}$$

Hence f_X is a valid PDF.

(ii) Calculation of $\mathrm{E}[X]$.

$$\begin{aligned} \mathrm{E}[X] &= \int_0^1 x\,6x(1-x)\,dx \\ &= 6\int_0^1 (x^2 - x^3)\,dx \\ &= 6\left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1 \\ &= 6\left(\frac{1}{3} - \frac{1}{4}\right) \\ &= \frac{1}{2}. \end{aligned}$$

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(iii) Calculation of $\mathrm{E}[X^2]$.

$$\begin{aligned}\mathrm{E}[X^2] &= \int_0^1 x^2\, 6x(1-x)\, dx \\ &= 6\int_0^1 (x^3 - x^4)\, dx \\ &= 6\left[\frac{x^4}{4} - \frac{x^5}{5}\right]_0^1 \\ &= 6\left(\frac{1}{4} - \frac{1}{5}\right) \\ &= \frac{3}{10}.\end{aligned}$$

(iv) Variance and standard deviation.

$$\begin{aligned}\mathrm{Var}(X) &= \mathrm{E}[X^2] - \Big(\mathrm{E}[X]\Big)^2 \\ &= \frac{3}{10} - \left(\frac{1}{2}\right)^2 \\ &= \frac{1}{20}.\end{aligned}$$

Therefore the standard deviation is

$$\sqrt{\mathrm{Var}(X)} = \sqrt{\frac{1}{20}} = \frac{1}{2\sqrt{5}}.$$

Hence,

$$\boxed{\mathrm{E}[X] = \frac{1}{2}, \quad \mathrm{E}[X^2] = \frac{3}{10}, \quad \mathrm{Var}(X) = \frac{1}{20}, \quad \sigma_X = \frac{1}{2\sqrt{5}}.}$$

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